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Full Length Article

Research on the dynamic response and disaster-resistance performance enhancement of hybrid tower under strong wind and earthquake

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ABSTRACT

With the rapid development of the wind power industry, wind turbine towers are evolving toward greater heights and larger capacities. When the tower height exceeds 140 m, hybrid towers demonstrate significant advantages over conventional steel towers in terms of economic efficiency and safety. Specifically, this hybrid system comprises a lower segmental concrete section compressed by external prestressed cables to prevent tension, a steel transition section, and an upper steel tower section. Although hybrid towers exceeding 160 m have been implemented in engineering practice, refined numerical simulations concerning their dynamic response under strong wind conditions and seismic actions remain limited. In this study, a 160 m-class hybrid tower located in a region with a seismic design intensity of 8 was selected as the prototype. Numerical models were established using ABAQUS to analyze the dynamic responses under strong wind conditions and seismic actions, respectively. Based on the Davenport fluctuating wind speed spectrum, a wind load time history corresponding to a wind speed of 37.5 m/s (strong wind conditions) was generated to analyze the wind-induced response. Furthermore, based on the design response spectrum for site type II, three ground motion records were selected from the PEER ground motion database. The seismic responses under frequent earthquake, design earthquake, and rare earthquake conditions were investigated, and measures to enhance the structural performance under rare earthquakes were proposed. The results indicate that the hybrid tower exhibits excellent performance under strong wind conditions, frequent earthquakes, and design earthquakes. No concrete tensile stress was observed in the concrete tower section, and the stress levels in key components—such as the upper steel tower section and prestressed cables—as well as the maximum tower top displacement, remained within allowable design limits. However, under rare earthquake action, concrete tensile stress developed in the concrete tower section, reaching a maximum value of 0.40 MPa. For the concrete tower section, the emergence of tensile stress indicates that the pre-compression stress at the concrete segment joint has been completely offset. This implies a potential opening of the segmented joints throughout the tower, thereby increasing the risk of collapse. Increasing the pre-tension force in the prestressed cables by 21.9% eliminated the tensile stress in the concrete tower section under rare earthquake action, ensuring the section remained in a compressive state throughout. Notably, even with this adjustment, the stresses in the upper steel tower section and prestressed cables, as well as the tower top displacement, remained within safe limits under strong wind conditions. The findings of this study provide a valuable reference for the safety evaluation of hybrid towers.

1. Introduction

Driven by the clean and stable nature of wind power generation, the industry is poised for rapid expansion in the coming decades. According to the Wind Energy Beijing Declaration 2.0 [1], China aims to achieve an installed wind power capacity of 5 billion kW by 2060, which represents a tenfold increase over the projected capacity for 2025. Consequently, onshore wind turbines are evolving toward larger scales. Turbines with hub heights of 180 m have already been connected to the grid, and wind turbine towers exceeding 200 m are currently under research and

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development. Notably, when the tower height exceeds 140 m, the hybrid tower has emerged as the dominant structural form due to its superior economic efficiency and mechanical performance. Typically, such a structure consists of a lower tower section made of precast concrete segments, a steel transition section, and an upper steel tower section. To counteract the tensile stresses induced by wind and seismic loads, external vertical prestressed cables are anchored from the foundation to the transition section, ensuring the concrete segments remain in a compressive state.

Onshore wind turbines are typically situated in high-altitude mountainous or hilly regions and are subjected to wind loads year-round. To investigate the responses and failure mechanisms of wind turbine towers under these conditions, numerous scholars have conducted extensive research. Chou et al. [2] analyzed the failure mechanisms of wind turbine towers under strong wind conditions, revealing that strong winds can induce buckling failure, leading to the formation of plastic hinges in the middle-to-upper sections of the tower and potentially causing overall collapse. Wu et al. [3] evaluated the wind loads on a 2 MW wind turbine structure based on the design code JG2010, showing good consistency between the design values and those obtained from GH simulations. Zhang et al. [4] investigated the dynamic response of wind turbine towers under wind loads, finding that the natural frequency of the tower is close to the fluctuating wind frequency, thereby increasing susceptibility to resonance. Dai et al. [5] examined the nonlinear response and collapse modes of wind turbine towers under typhoon conditions. Ding et al. [6] analyzed the response of bucket foundations for offshore wind turbines under wind loads, observing that continuous loading affects the soil stress level and induces a distinct saddle-shaped settlement zone. Li et al. [7] established a joint probability distribution model of wind speed and direction to investigate their effects on the fatigue life of wind turbine towers. Jing et al. [8] proposed a joint optimization strategy involving data interaction between finite element software and topology optimization programs for the design and wind-induced response analysis of a 5 MW wind turbine tower. Jaimes et al. [9] proposed a probabilistic risk assessment method to predict the economic impact of hurricanes on wind turbines, highlighting the importance of the selected indicators. Kelele et al. [10] employed time- and frequency-domain simulation methods to comparatively analyze the dynamic response of wind turbine towers under turbulent wind loads, indicating that accuracy varies with sample size. Bao et al. [11] explored a novel active pitch control method to regulate blade loads, confirming its reliability in modulating blade dynamic responses. Xu et al. [12] developed an integrated fully coupled model for onshore wind turbines with steel–concrete hybrid towers and conducted linear and nonlinear dynamic response analyses under extreme wind conditions. Bazilevs et al. [13] investigated the aerodynamic characteristics of wind turbine rotors based on CFD methods, demonstrating a significant discrepancy between the actual power of the blades and that calculated under the assumption of rigid blades.

When wind turbine towers are located in high seismic intensity regions, their responses under rare earthquake actions cannot be ignored. Consequently, numerous scholars have conducted relevant research to investigate the seismic performance of these structures. Santangelo et al. [14,15] analyzed the seismic performance of onshore and offshore wind turbines under seismic actions, respectively, and investigated the associated levels of aerodynamic damping. Du et al. [16] investigated the dynamic response of offshore wind turbines under seismic loads and found that the peak acceleration of the tower occurs at the mid-height section. Chen et al. [17] studied the seismic response of offshore wind turbine foundations using OpenSees software, indicating that the tower undergoes plastic deformation under ground motions. Xu et al. [18] conducted shaking table tests to investigate the effects of near-fault pulse-like ground motions on the seismic response of wind turbine towers, demonstrating that such motions significantly influence the structural response. Ma et al. [19] examined the tower response under seismic actions and discovered that foundation spring stiffness significantly affects the foundation bending moment and natural frequency. Wen et al. [20] analyzed the dynamic response of wind turbine towers under various seismic waves, revealing that certain members sustain plastic damage under rare earthquake actions. Zhao et al. [21] analyzed the response of wind turbines under seismic waves with varying spectral characteristics and found that short-period ground motions can induce full-section plastic hinges in the middle-to-upper parts of the tower. Dai et al. [22] analyzed the response of a 1.5 MW wind turbine under ground motions with distinct spectral characteristics, indicating that long-period ground motions significantly amplify the tower top displacement and acceleration responses. Dai et al. [23] proposed and validated a seismic analysis method for wind turbine towers. Serras et al. [24] quantified the performance of traditional steel wind turbine towers under repeated seismic actions, observing that such actions significantly affect their seismic behavior. Liu et al. [25] analyzed the dynamic response of wind turbine towers under seismic loads, showing significant stress concentration at the tower top opening when the direction of the seismic load is perpendicular to that of the wind load. Shen et al. [26] derived conversion relationships among seismic intensity, ground acceleration, and horizontal seismic influence coefficients for wind turbine towers with different design service lives to meet seismic fortification requirements. Song et al. [27] investigated the influence of near-fault ground motions on the seismic response of wind turbine towers and analyzed the effects of velocity pulses and vertical ground motions on the structural collapse mechanism. Mohamed et al. [28] evaluated the seismic performance of offshore platforms and proposed retrofitting methods. Wang et al. [29] studied the impact of Tuned Mass Dampers (TMD) on the seismic mitigation performance of steel–concrete hybrid towers, indicating that optimizing TMD parameters can effectively enhance the control performance of the tower. Xu et al. [30] investigated the imperfection sensitivity of wind turbines subjected to near-fault pulse-like seismic records, showing that imperfections influence the dispersion of the seismic response. Kapasakalis [31] examined the effects of various vibration control systems on the seismic performance of wind turbines, demonstrating that these systems can effectively control the seismic response. Ngo et al. [32] evaluated the reliability of simplified spring models in assessing seismic performance, indicating that such models allow for rapid assessment. Mellas et al. [33] investigated the seismic response of wind turbines considering the SSI effect based on simulations and scaled model tests, showing that soil characteristics significantly affect seismic performance. Chen et al. [34] explored the application of inerter-based passive structural control for suppressing the seismic response of wind turbines, indicating that this approach improves the dynamic response of the tower more effectively than traditional TMDs.

With the rapid development of the wind power industry, wind turbine towers are evolving toward greater heights and larger capacities. Due to advantages such as economic efficiency and superior mechanical performance, hybrid towers are gradually becoming the dominant structural form. Although hybrid towers exceeding 160 m have been applied in engineering practice, in-depth research on their disaster-resistance performance under strong wind conditions and seismic actions remains insufficient [35,36]. The structural system of a hybrid tower is highly complex, comprising the upper steel tower section, the transition section, and the concrete tower section. Additionally, prestressed cables extending from the foundation to the lower part of the transition section are required to ensure the concrete tower section remains in a compressive state. Currently, refined numerical simulations targeting this specific structural system—particularly key components such as the concrete tower section, prestressed cables, and the upper steel tower section—are inadequate [37,38]. Existing studies predominantly utilize simplified models to represent wind turbine structures. For instance, Xing et al. [39] simplified the blades and nacelle into a lumped mass positioned at the tower top. However, this approach neglects the dynamic impact of eccentric moments on the structure. Furthermore, in the simulation of connections for the concrete tower section, some

researchers rely solely on simple tie constraints to model the interface between the segments and the structural adhesive. Such simplifications fail to capture the complex contact states inherent in practical engineering projects. To investigate the mechanical performance of hybrid towers under strong wind conditions and seismic actions, a 160 m-class hybrid tower was selected as a prototype. A finite element model was established using ABAQUS software to conduct dynamic time-history analyses under strong wind conditions and seismic actions, respectively, thereby examining the stress states of various components. To address the safety hazards identified in the concrete tower section under rare earthquake actions, measures to enhance disaster-resistance performance were proposed and validated through numerical simulations.

Regarding the details of the finite element modeling, a comprehensive full solid model of the wind turbine structure was established. The hub and three blades were discretized using C3D10 tetrahedral elements, comprising a total of 1654 elements. To simulate the connections between the concrete tower sections, tie constraints were employed to achieve inter-segment interactions based on the engineering drawings. This refined model incorporates 42 concrete segments, the transition section, the upper steel tower section, 16 prestressed cables, anchor devices, and the turbine unit. All parameters remain consistent with the actual engineering project, and constraints were strictly applied following the practical construction sequence.

2. Project overview

In this study, a 160 m-class hybrid tower was selected as the prototype. The project site is characterized by a basic wind speed of 37.5 m/s. In this study, the basic wind speed is taken as the 10-min mean wind speed at 10 m above ground level with a 50-year return period, consistent with the definition of the reference wind speed $\overline{V}_{10}$ adopted in the Davenport spectrum. The site condition is classified as site type II, with a seismic precautionary intensity of 8 degrees. The hybrid tower comprises three primary components: the concrete tower section, the transition section, and the upper steel tower section, with a total structural height of 163.24 m. Specifically, the concrete tower section consists of 42 concrete segments, each with a height of 3.08 m, resulting in a total section height of 129.36 m. This section features a tapered geometry, with top and bottom diameters of 5.924 m and 11.15 m, respectively, and a uniform wall thickness of 0.3 m. The transition section has a height of 3.08 m, a constant diameter of 5.924 m, and a wall thickness of 0.02 m. The upper steel tower section stands 30.8 m tall, tapering from a bottom diameter of 5.924 m to a top diameter of 4.644 m, with a wall thickness of 0.02 m. The anchor device has a diameter of 0.3 m and a thickness of 0.15 m. The nacelle dimensions are 8.52 $m \times 5.26\ m \times 5.26$ m. The turbine utilizes three blades, each with a radius of 100 m, connected to a hub with a radius of 4.5 m. To ensure structural stability, the tower is equipped with 16 bundles of prestressed cables, each composed of twenty-one 1×7 standard Φ15.2 steel strands. Regarding material properties, the concrete segments are fabricated from C80 concrete, while the transition section, the upper steel tower section, and the anchor devices are constructed from Q345 steel. The reinforcement grade is HRB400, and the blades are made of glass fiber-reinforced resin. The prestressed cables utilize low-relaxation high-strength steel strands with a standard characteristic ultimate strength of 1860 MPa and a tension control stress of 1000 MPa. The geometric configuration and dimensions of the hybrid tower are illustrated in Fig. 1, and the detailed parameters are listed in Table 1.

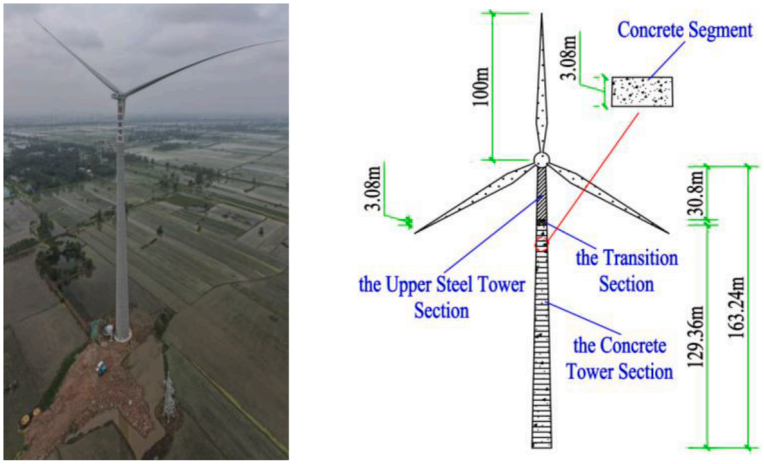
3. Finite element model

3.1. Model establishment

The finite element model of the hybrid tower was established using ABAQUS software. The concrete tower section, the transition section, the upper steel tower section, and the anchor devices were discretized using C3D8R hexahedral elements, with element counts of 31,920, 2884, 11,700, and 1268, respectively. The hub and three blades were modeled using C3D10 tetrahedral elements, totaling 1654 elements, while the prestressed cables were simulated using T3D2 truss elements. For the contact interfaces between adjacent concrete segments in the concrete tower section, "Hard" contact behavior was defined in the normal direction, and a "Penalty" friction formulation with a coefficient

Table 1
Parameters of the hybrid tower.

Section	Parameter	Value (m)	Material
Concrete Tower Section	Height	129.36	C80 Concrete
	Top Diameter	5.924	
	Bottom Diameter	11.15	
	Wall Thickness	0.3	
Transition Section	Height	3.08	Q345 Steel
	Top Diameter	5.924	
	Bottom Diameter	5.924	
	Wall Thickness	0.02	
Upper Steel Tower Section	Height	30.8	Q345 Steel
	Top Diameter	4.644	
	Bottom Diameter	5.924	
	Wall Thickness	0.02	



(a) Appearance of the Hybrid Tower (b) Dimensions of the Hybrid Tower

Fig. 1. The 160 m hybrid tower.

of 0.6 was applied in the tangential direction. Tie constraints were applied at the interfaces between the concrete tower section and the transition section, as well as between the transition section and the upper steel tower section. The reinforcement cage was simulated using the "Embedded region" constraint within the concrete tower section. A coupling constraint was established between the upper endpoints of the prestressed cables and the top surface of the anchor devices. Additionally, tie constraints were employed to connect the tower to the nacelle, and the nacelle to the blades, assuming no relative slippage occurs. This modeling approach ensured the accuracy of the mass and stiffness matrices of the hybrid tower. The established finite element model is illustrated in Fig. 2.

3.2. Boundary conditions and material constitutive models

Boundary conditions significantly influence the mechanical performance and dynamic response in finite element analysis. As illustrated in Fig. 3, fully fixed constraints were applied to the base of the hybrid tower and the lower endpoints of the prestressed cables. These constraints restrained nodal displacements and rotations in all degrees of freedom, effectively simulating the connection between the hybrid tower and the foundation. Consequently, this configuration ensured that

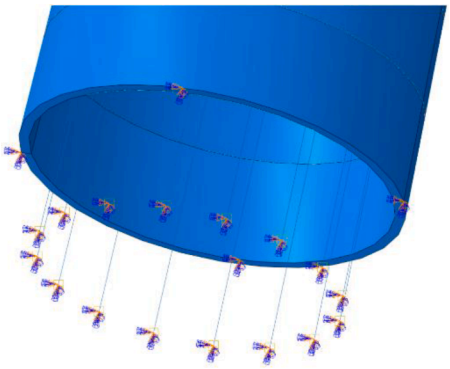
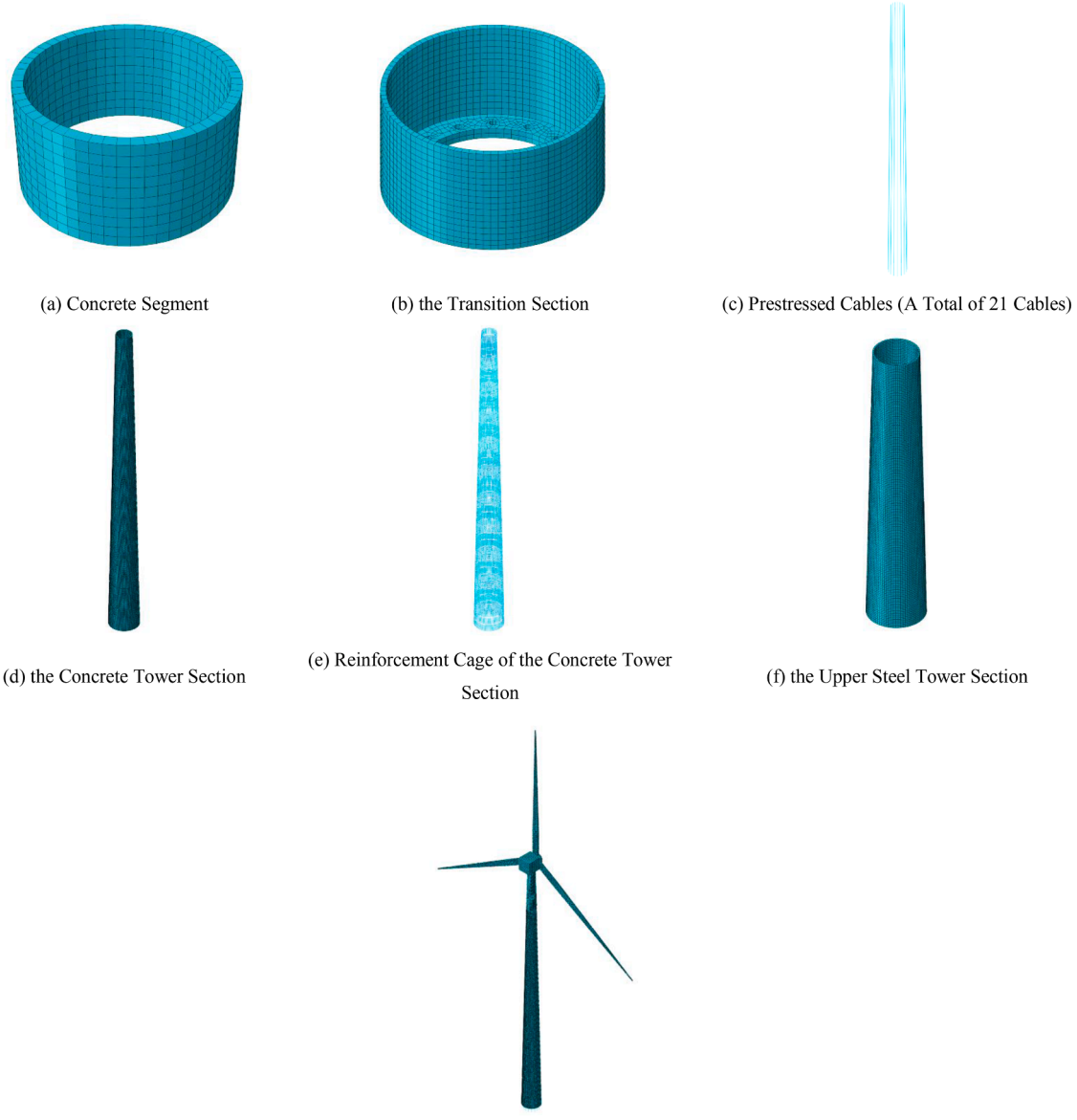


Fig. 3. Viscoelastic boundary.



(a) Concrete Segment
(b) the Transition Section
(c) Prestressed Cables (A Total of 21 Cables)
(d) the Concrete Tower Section
(e) Reinforcement Cage of the Concrete Tower Section
(f) the Upper Steel Tower Section
(f) Full Finite Element Model

Fig. 2. Finite Element Model.

the model accurately reflected the actual mechanical state of the hybrid tower under loading conditions.

A fully fixed base is adopted to provide a consistent reference condition for the comparative analyses. It is acknowledged that soil–structure interaction may affect modal properties and rare-earthquake demand; thus, the concrete tensile-stress conclusions in this paper are conditioned on the fixed-base assumption.

The concrete tower section adopts the Concrete Damaged Plasticity (CDP) constitutive model to capture the nonlinear behavior of high-strength concrete under compression and tension. Both the transition section and the upper steel tower section are composed of Q345 steel, governed by an elastic-plastic constitutive model. Additionally, Rayleigh damping is incorporated to accurately characterize the dynamic properties of the structure. The reinforcement cage consists of HRB400 grade steel and similarly adopts an elastic-plastic constitutive model. The prestressed cables, made of low-relaxation high-strength steel strands, are also modeled using an elastic-plastic constitutive relation. Specifically, the linear thermal expansion coefficient of the prestressed cables is defined to facilitate the application of prestress via the equivalent cooling method. The constitutive relations and relevant parameters for the main components of the hybrid tower are listed in Table 2.

3.3. Application of prestress

In the finite element modeling process, the accurate application of prestress is crucial for reproducing the global mechanical performance of the hybrid tower and ensuring consistency with the actual stress state. In this study, the equivalent cooling method was employed within ABAQUS to introduce prestress into the prestressed cables. By subjecting the prestressed cables to a negative temperature load, this method utilizes the linear thermal expansion properties of the material to generate an axial tensile force equivalent to the actual tensioning effect, thereby effectively establishing the prestress. Compared to the direct application of initial stress or displacement, the equivalent cooling method not only mitigates convergence issues during numerical calculations but also more accurately captures the mechanism of prestress transfer under loading conditions. The specific equivalent cooling value is calculated using Eq. (1).

$$\Delta T = \frac{F}{\alpha EA} = \frac{\sigma}{\alpha E} \tag{1}$$

Where ΔT represents the equivalent cooling value, F denotes the pretension value, σ is the prestress value, A is the cross-sectional area of the prestressed cables, E is the elastic modulus of the prestressed cables, and α is the linear thermal expansion coefficient of the prestressed cables. In this study, a prestress of 1000 MPa was applied to the prestressed cables. Based on Eq. (1), the required equivalent cooling value is calculated to be 427 °C.

Table 2
Constitutive materials and parameters.

Section	Constitutive Model	Parameter	Value
Concrete Tower Section	CDP	Density / kg·m^{-3}	2650
		Elastic Modulus / GPa	38
		Poisson's Ratio	0.2
		Tensile Yield Stress / MPa	3.9
		Compressive Yield Stress / MPa	80
		Dilation angle	38°
		Eccentricity	0.1
		Ratio of biaxial to uniaxial compressive strength	1.16
		Second stress invariant ratio	0.667
		Viscosity parameter	0.0005
Transition Section and Upper Steel Tower Section	Elastic–Plastic	Density / kg·m^{-3}	7500
		Elastic Modulus / GPa	200
		Poisson's Ratio	0.3
		Yield Stress / MPa	345
		Rayleigh Damping Coefficient α	0.07
		Rayleigh Damping Coefficient β	0.01
Reinforcement Cage	Elastic–Plastic	Density / kg·m^{-3}	7800
		Elastic Modulus / GPa	200
		Poisson's Ratio	0.3
		Yield Stress / MPa	400
Prestressed Cables	Elastic–Plastic	Density / kg·m^{-3}	7850
		Elastic Modulus / Gpa	195
		Poisson's Ratio	0.3
		Yield Stress / MPa	1860
		Linear Thermal Expansion Coefficient	0.000012

3.4. Wind load calculation

Wind hazard constitutes one of the primary threats affecting the dynamic response and safe operation of tall, flexible structures. For hybrid towers situated in complex wind environments, wind load not only governs the accumulation of fatigue damage during the service life but also directly impacts structural safety and operational reliability under extreme conditions. To evaluate the safety performance of the hybrid tower under wind hazard, this study calculates the wind loads acting on the structure at a wind speed of 37.5 m/s, based on the Davenport fluctuating wind speed spectrum.

Fluctuating wind is a critical factor inducing structural vibration, and the relevant design parameters are determined in accordance with applicable specifications. The fluctuating wind pressure is expressed by Eq. (2) as:

$$P(h_i, t) = I_0(h_i)B(h_i)u(t) \tag{2}$$

Where, P(hi, t) denotes the fluctuating wind pressure at height h_i above the ground at time t; $I_0(h_i)$ represents the spatial correlation of fluctuating wind pressure, which is a random variable with a variance of 1; u(t) constitutes the wind speed spectrum; $B(h_i)$ is the wind pressure intensity coefficient, given by Eq. (3):

$$B\left(h_i\right) = A_i \mu_s\left(h_i\right)\mu_z\left(h_i\right)\omega_0\sqrt{\frac{24K_r}{\mu_z(h_i)}} \tag{3}$$

In the formula: A_i, $\mu_s(h_i)$, $\mu_z(h_i)$, ω_0 are the windward area of the structure at the height h_i above the ground, the wind load shape coefficient, the wind pressure height change coefficient and the basic wind pressure; K_r is related to the roughness of the ground.

To account for the variation of wind speed with height, the vertical mean wind profile is modeled using the power law, expressed as:

$$\overline{v}\left(z\right) = V_{10}\left(\frac{z}{10}\right)^{\beta} \tag{4}$$

Where $\overline{v}(z)$ is the mean wind speed at height z; V_{10} is the basic wind speed at the reference height of 10 m; and β is the terrain roughness exponent, taken as 0.15 for site type II. This profile ensures physical consistency with the boundary layer characteristics.

The power spectral density function of the Davenport wind speed spectrum is represented by Eq. (5) as:

$$S_u\left(\omega\right) = \frac{4}{3}\pi\frac{a(z)^2\omega}{(1+a(z)^2\omega^2)^{\frac{4}{3}}} \tag{5}$$

In the formula: $\omega \geq 0$; $a\left(z\right) = \frac{600}{\pi\overline{v}(z)}$; $\overline{v}(z)$ represents the standard

average wind speed at a height of z m above the ground. A MATLAB program was developed to calculate the Davenport fluctuating wind speed spectrum. To accurately determine the wind loads acting on the hybrid tower, the upper steel tower section was divided into 10 segments, while the blades and hub were collectively discretized into 7 segments. The concrete tower section, consisting of 42 individual concrete segments, required no further subdivision. The segmentation scheme of the hybrid tower for load calculation is illustrated in Fig. 4. Using the developed program, wind load time histories for each segment were generated at a wind speed of 37.5 m/s, with a time step of 0.1 s and a total duration of 100 s. The duration of 100 s was selected to ensure that the simulated response covers sufficiently many cycles of the low-frequency global modes of the tower (i.e., on the order of tens of fundamental periods), while maintaining a reasonable computational cost for the subsequent nonlinear time-history analyses. The present wind-response results are therefore intended primarily for comparative assessment among different hazard cases and prestress schemes. It is acknowledged that a longer realization (e.g., several hundred seconds or more) would provide more statistically stable estimates of peak and RMS responses for stochastic wind loading; thus, the reported peaks may exhibit some sample-length dependence, which is a limitation of the current study.

These wind load time histories were subsequently input into the ABAQUS model to simulate the dynamic response of the hybrid tower under wind hazard. As examples, the wind load time histories for the top concrete segment, the transition section, Segment 1 of the upper steel tower section, and Segment 1 of the blade are presented in Fig. 5.

Aerodynamic damping is not explicitly considered in the present wind-load analysis; therefore, the wind-induced responses may be conservative in amplitude for operating wind turbines. This simplification is adopted due to the lack of aeroelastic/control parameters and is treated as a limitation of this study.

3.5. Selection of seismic waves

The Pacific Earthquake Engineering Research Center (PEER) database offers a comprehensive collection of earthquake records. To evaluate the safety performance of the hybrid tower under frequent earthquake, design earthquake, and rare earthquake conditions, three seismic waves were selected from the PEER database based on the design response spectrum for site type II. The details of the selected seismic waves are listed in Table 3, and a comparison between their response spectra and the standard design response spectrum is illustrated in Fig. 6.

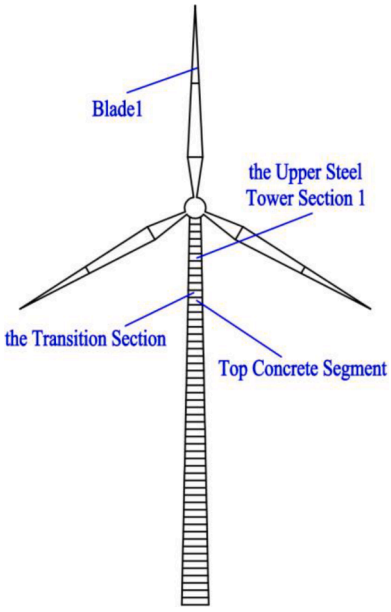


Fig. 4. Segmentation diagram of the hybrid tower for wind load calculation.

To comprehensively evaluate the safety performance of the hybrid tower under varying seismic intensities, the peak accelerations of the three selected seismic waves were scaled to 0.07 g (frequent earthquake), 0.20 g (design earthquake), and 0.40 g (rare earthquake), respectively. These seismic waves have a time step of 0.02 s. To enhance computational efficiency, the first 18 s of each wave were selected as the analysis duration. After extracting the first 18 s, each record was minimally baseline-corrected (mean removal and linear detrending) to ensure near-zero residual velocity and displacement at the end of the truncated segment. No filtering was applied to preserve low-frequency content; residual checks were negligible, so the influence on key stress results is expected to be limited. The time histories of the three seismic waves are presented in Fig. 7.

4. Disaster-resistance performance analysis of hybrid towers under wind hazard

To evaluate the structural safety and operational reliability of the hybrid tower under wind hazards, this study investigated the dynamic response under a basic wind speed of 37.5 m/s. The disaster-resistance performance of the hybrid tower was comprehensively assessed by analyzing the stress responses of the concrete tower section, the transition section, the upper steel tower section, and the prestressed cables, as well as the tower top displacement response. The time histories of stresses in the concrete tower section, the transition section, the upper steel tower section, and the prestressed cables, along with the tower top displacement time history, are presented in Fig. 8. The results indicate that the hybrid tower exhibits excellent overall disaster-resistance performance under strong wind conditions.

The overall stress levels of the hybrid tower under strong wind conditions remain within a safe range. As illustrated in Fig. 8(a), the concrete tower section exhibits compressive stress throughout the entire time history, with no occurrence of concrete tensile stress. The maximum concrete compressive stress is 13.74 MPa, which is significantly lower than the design compressive strength of C80 concrete. This indicates that there is no risk of tensile cracking in the concrete tower section under strong wind conditions; furthermore, the concrete compressive stress remains at a low level with an adequate safety margin, verifying the reliability of the prestress design. The maximum stresses in the transition section and the upper steel tower section are 121.41 MPa and 75.35 MPa, respectively, both of which are markedly lower than the yield strength of Q345 steel. This suggests that the steel structure—as the critical connection between the concrete tower section and the nacelle—remains in the elastic stage under strong wind conditions, possessing a sufficient safety reserve to effectively resist wind-induced bending moments and shear forces. The initial tension of the prestressed cables is 1000 MPa; under wind loads, the maximum stress increases to 1012.31 MPa. The stress increment is only 12.31 MPa, leaving a substantial margin relative to the ultimate strength of the steel strands. This demonstrates that the prestressed cables remain stable under high wind action, continuously providing effective constraint to the concrete tower section, thereby enhancing the bending stiffness and integrity of the overall structure.

Tower top displacement is a crucial indicator of overall structural stiffness and stability. As shown in Fig. 8(e), under a wind load corresponding to 37.5 m/s, the maximum tower top displacement is 0.57 m, representing approximately 0.35 % of the total tower height. For such super-tall flexible structures, this deformation falls within conventional design limits and poses no threat to the normal operation or structural safety of the hybrid tower. The displacement time-history curve exhibits stable periodic oscillation without any divergent trend, indicating that the structure possesses favorable damping characteristics capable of effectively dissipating wind energy input and preventing aerodynamic

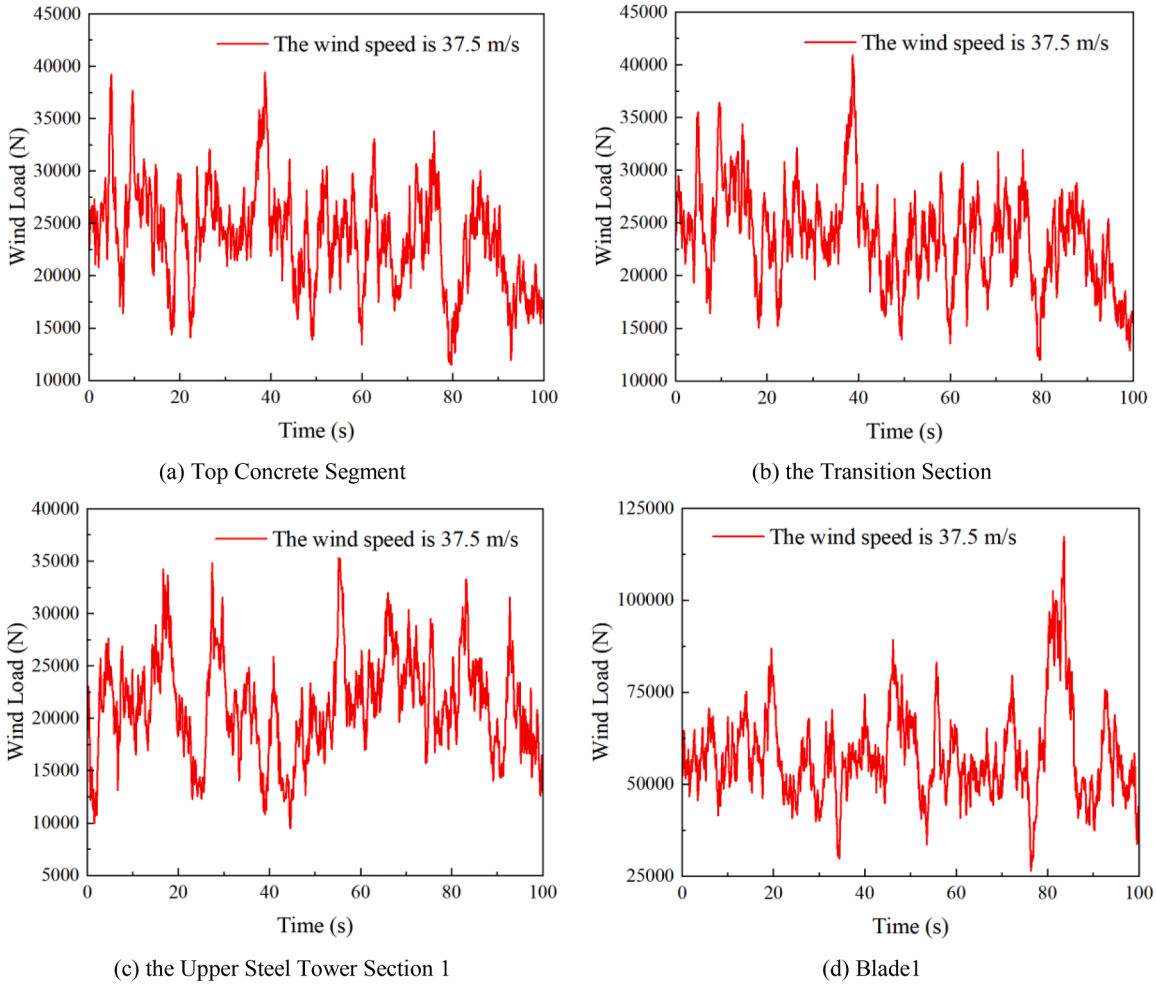


Fig. 5. Dynamic Time History of Wind Load.

Table 3
Selected earthquake waves.

Seismic Wave Name	Recording Location	Magnitude	Peak Acceleration /g	Predominant Period /s
Humbolt Bay (1937)	Ferndale City Hall	6.90	0.200	0.21
Helena, Montana-01(1935)	Carroll College	7.51	0.204	0.24
Helena, Montana-02(1935)	Helena Fed Bldg	7.51	0.181	0.20

instability.

In summary, the hybrid tower demonstrates excellent disaster-resistance performance under wind hazards. Under the basic wind speed condition, the stresses in all critical components are significantly lower than the material strength limits, ensuring a sufficient safety margin. Simultaneously, the overall structural displacement is controlled within a reasonable range. The analysis results effectively demonstrate the validity of the hybrid tower structural design, confirming its superior disaster-resistance performance and structural reliability in withstanding strong wind hazards.

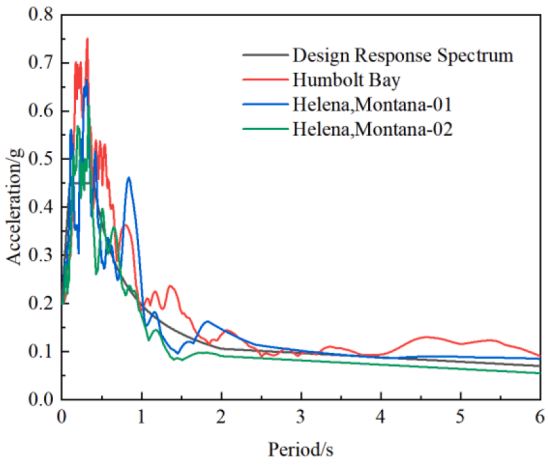


Fig. 6. Comparison between the response spectra of the selected seismic waves and the design response spectrum.

5. Disaster-resistance performance analysis of hybrid tower under earthquake hazard

To evaluate the disaster-resistance performance of the hybrid tower under varying seismic intensities, three seismic waves were selected from the Pacific Earthquake Engineering Research Center (PEER)

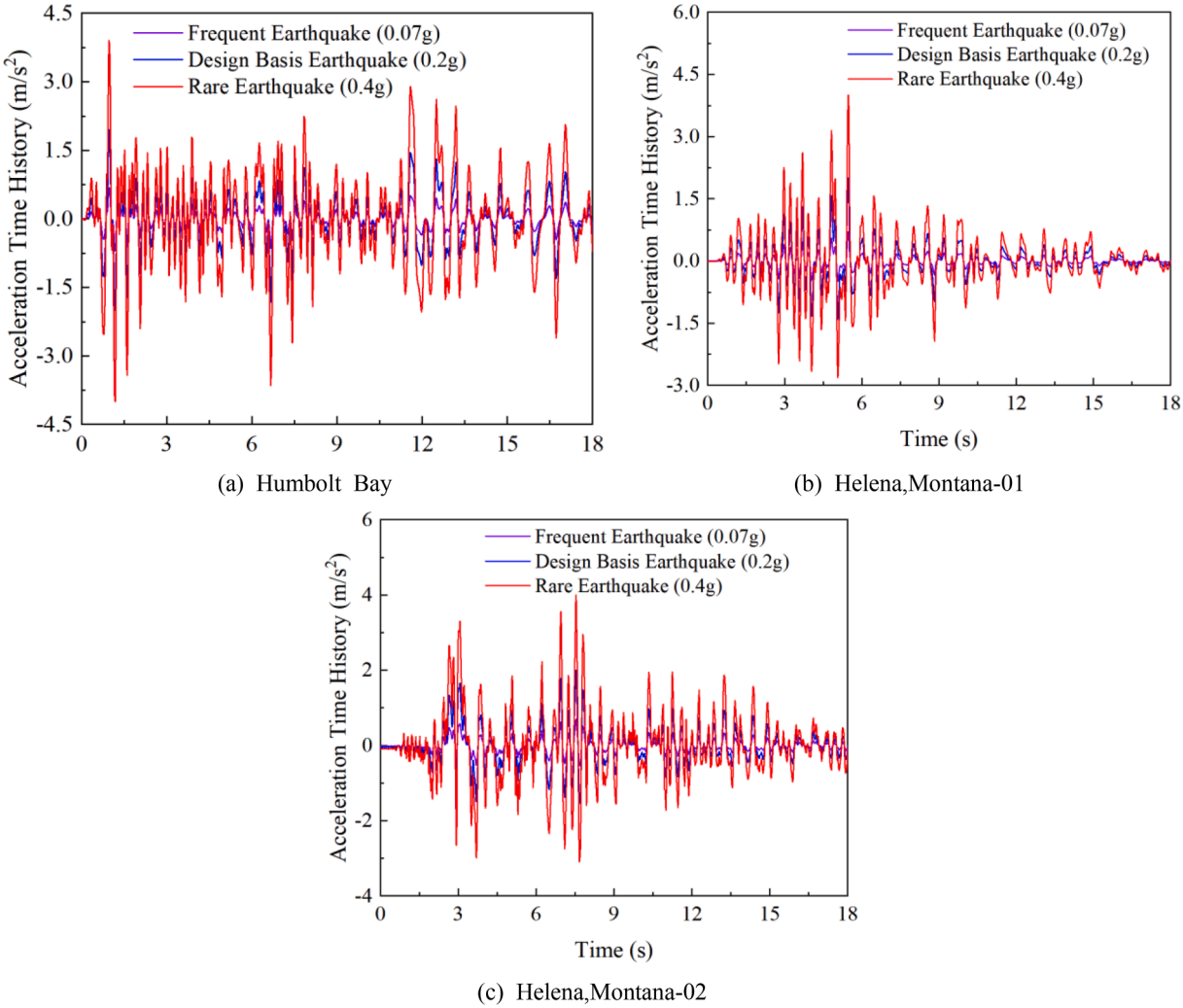


Fig. 7. Scaled seismic waves after amplitude modulation.

database. Their peak accelerations were scaled to 0.07 g, 0.20 g, and 0.40 g, corresponding to frequent earthquake, design earthquake, and rare earthquake levels, respectively. Based on these inputs, the seismic behavior of the hybrid tower was investigated. The disaster-resistance performance under varying seismic hazards was comprehensively evaluated by analyzing the stress levels of critical components—including the concrete tower section, the transition section, the upper steel tower section, and the prestressed cables—as well as the tower top displacement response.

5.1. *Disaster-resistance performance under frequent earthquakes*

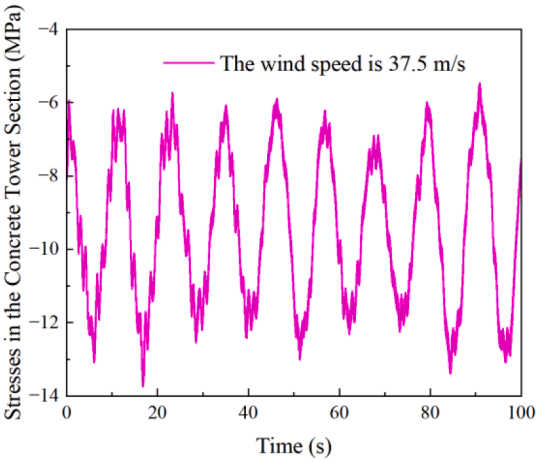
The time histories of stresses in the concrete tower section, the transition section, the upper steel tower section, and the prestressed cables, along with the tower top displacement time history under frequent earthquakes, are presented in Fig. 9. The results indicate that the hybrid tower exhibits excellent overall disaster-resistance performance under frequent earthquake conditions.

Under the excitation of the Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves, the stress in the concrete tower section remained compressive throughout the entire time history, with no occurrence of tensile stress. The maximum concrete compressive stresses were 9.79 MPa, 9.90 MPa, and 10.33 MPa, respectively. These values are significantly lower than the design compressive strength of C80 concrete, indicating no risk of tensile cracking in the concrete tower section under frequent earthquakes. Furthermore, the concrete compressive stress remains at a low level with a sufficient safety margin, thereby validating the reliability of the prestress design.

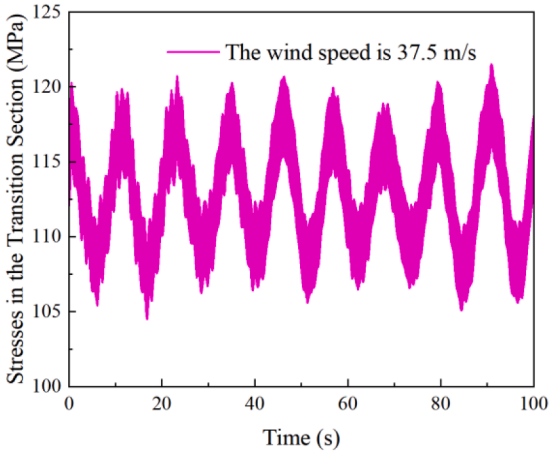
Regarding the steel components, the maximum stresses in the transition section under the three seismic waves were 120.37 MPa, 120.67 MPa, and 122.88 MPa, respectively. Similarly, the maximum stresses in the upper steel tower section were 76.42 MPa, 76.14 MPa, and 80.05 MPa, respectively. Both sets of values are substantially lower than the yield strength of Q345 steel. This demonstrates that the steel structure components remain within the elastic range under frequent earthquakes, possessing a high seismic safety margin.

The initial tension of the prestressed cables is 1000 MPa. Under the specified seismic waves, the maximum stresses increased to 1000.88 MPa, 1001.34 MPa, and 1001.85 MPa, respectively. The stress increments were merely 0.088 %, 0.13 %, and 0.19 %, leaving a substantial margin relative to the ultimate strength of the steel strands. This indicates that the prestressed cables maintain a stable constraining effect under frequent earthquakes.

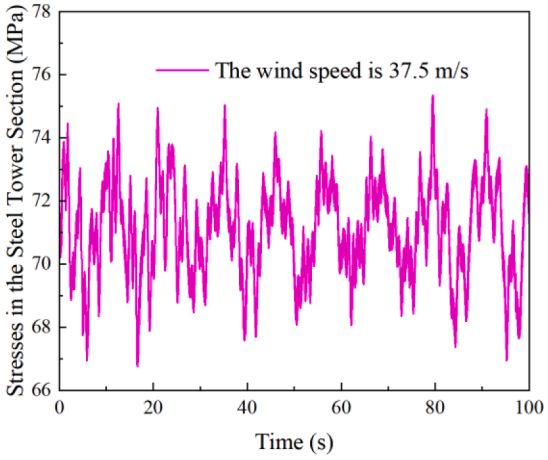
Tower top displacement serves as a critical indicator of overall structural stiffness and stability. Under the Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves, the tower top displacements were 0.53 m, 0.50 m, and 0.54 m, respectively, corresponding to 0.32 %, 0.31 %, and 0.33 % of the total tower height. For such super-tall flexible structures, these deformations fall within conventional design limits, suggesting that the overall deformation of the



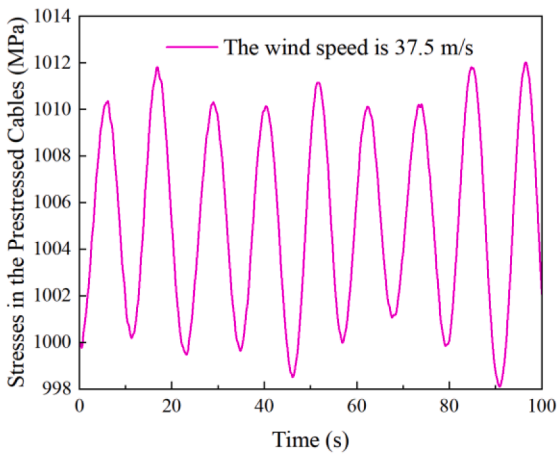
(a) Stresses in the Concrete Tower Section



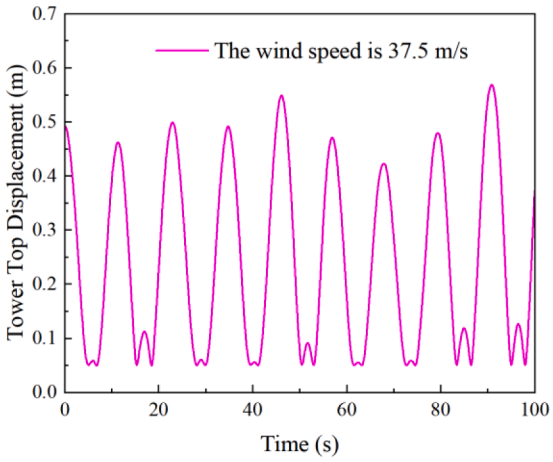
(b) Stresses in the Transition Section



(c) Stresses in the Steel Tower Section



(d) Stresses in the Prestressed Cables



(e) Tower Top Displacement

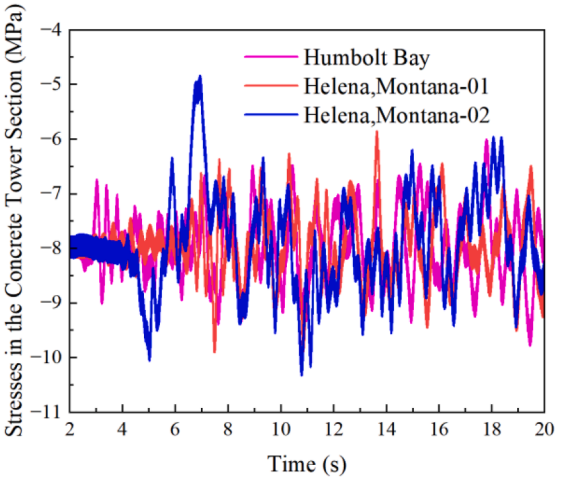
Fig. 8. Dynamic response of the hybrid tower under wind hazard.

hybrid tower is well-controlled under frequent earthquakes, posing no threat to operational safety or structural stability.

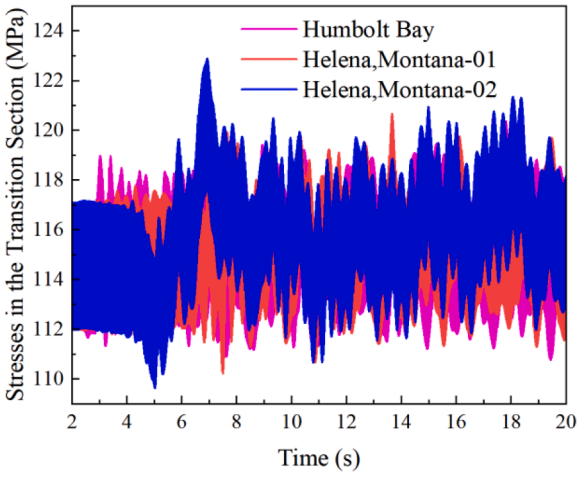
In summary, under frequent earthquake actions, the concrete tower section, steel structure sections, and the prestressed cables of the hybrid tower all exhibit favorable mechanical states. Additionally, the tower top displacement remains within a reasonable range. The overall disaster-resistance performance is excellent, ensuring the safety and normal operation of the structure under this condition.

5.2. *Disaster-resistance performance under design earthquakes*

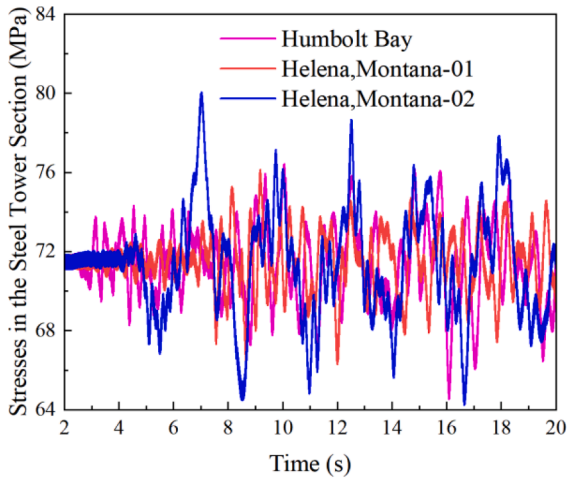
The time histories of stresses in the concrete tower section, the



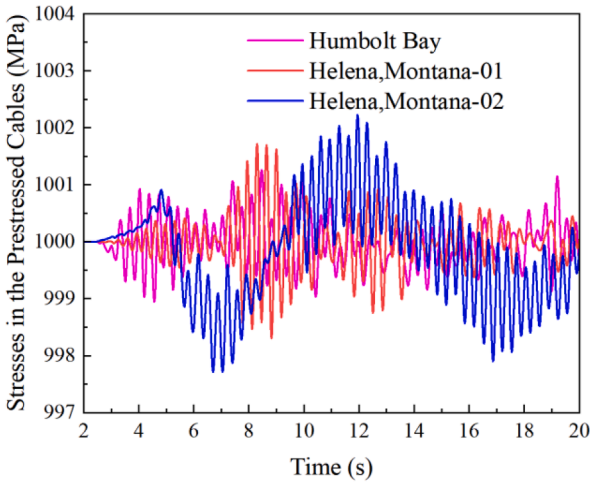
(a) Stresses in the Concrete Tower Section



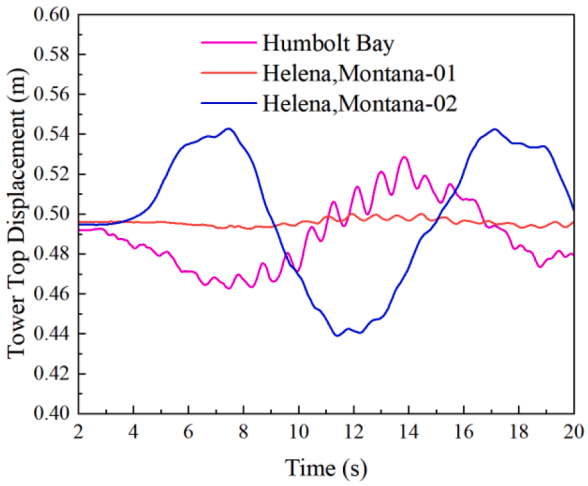
(b) Stresses in the Transition Section



(c) Stresses in the Steel Tower Section



(d) Stresses in the Prestressed Cables

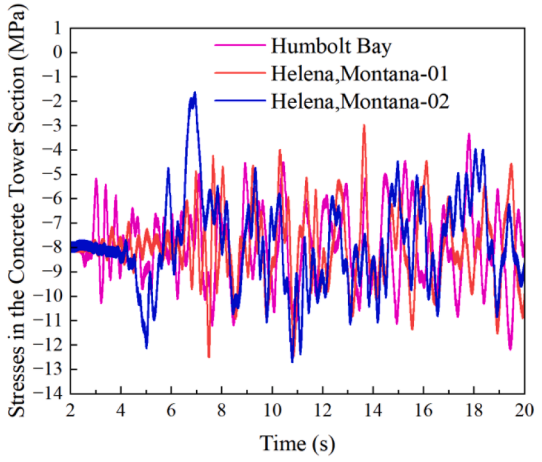


(e) Tower Top Displacement

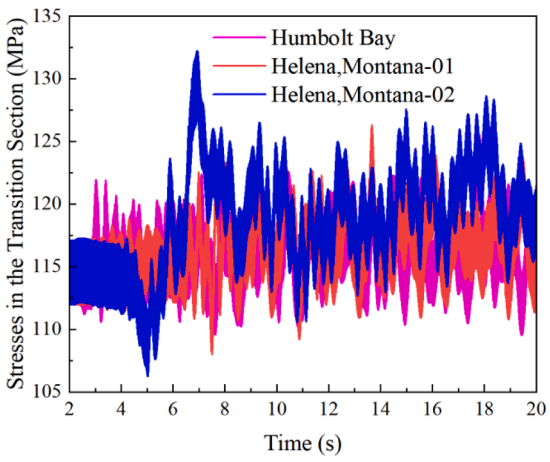
Fig. 9. Dynamic response of the hybrid tower under frequent earthquake.

transition section, the upper steel tower section, and the prestressed cables, along with the tower top displacement time history under design earthquakes, are presented in Fig. 10. The results indicate that the hybrid tower exhibits overall favorable disaster-resistance performance under design earthquake conditions.

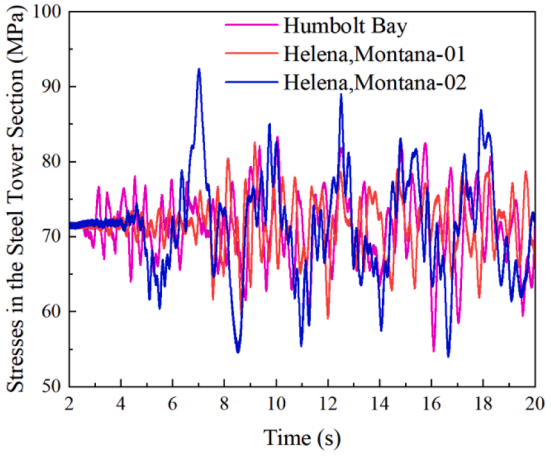
Under the excitation of the Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves, the stress in the concrete tower section remained compressive throughout the entire time history, with



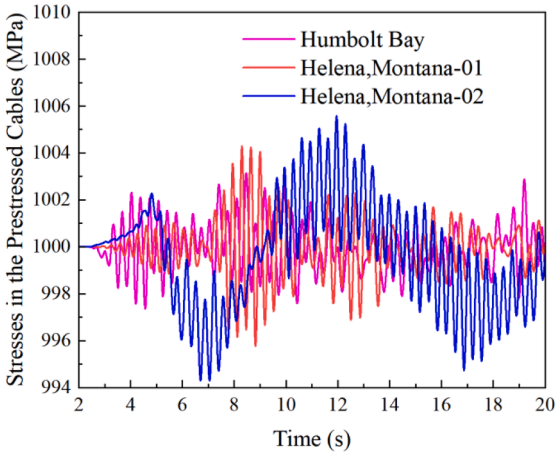
(a) Stresses in the Concrete Tower Section



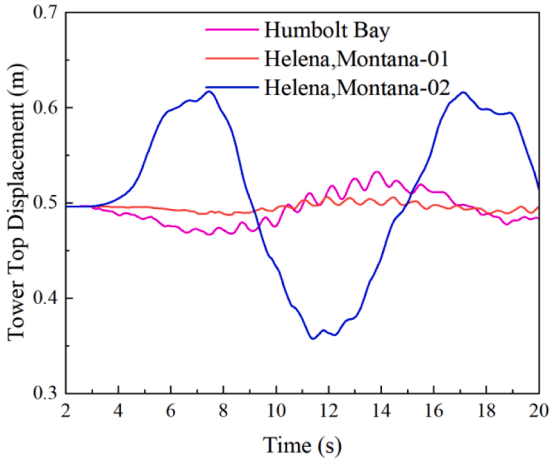
(b) Stresses in the Transition Section



(c) Stresses in the Steel Tower Section



(d) Stresses in the Prestressed Cables



(e) Tower Top Displacement

Fig. 10. Dynamic response of the hybrid tower under design earthquake.

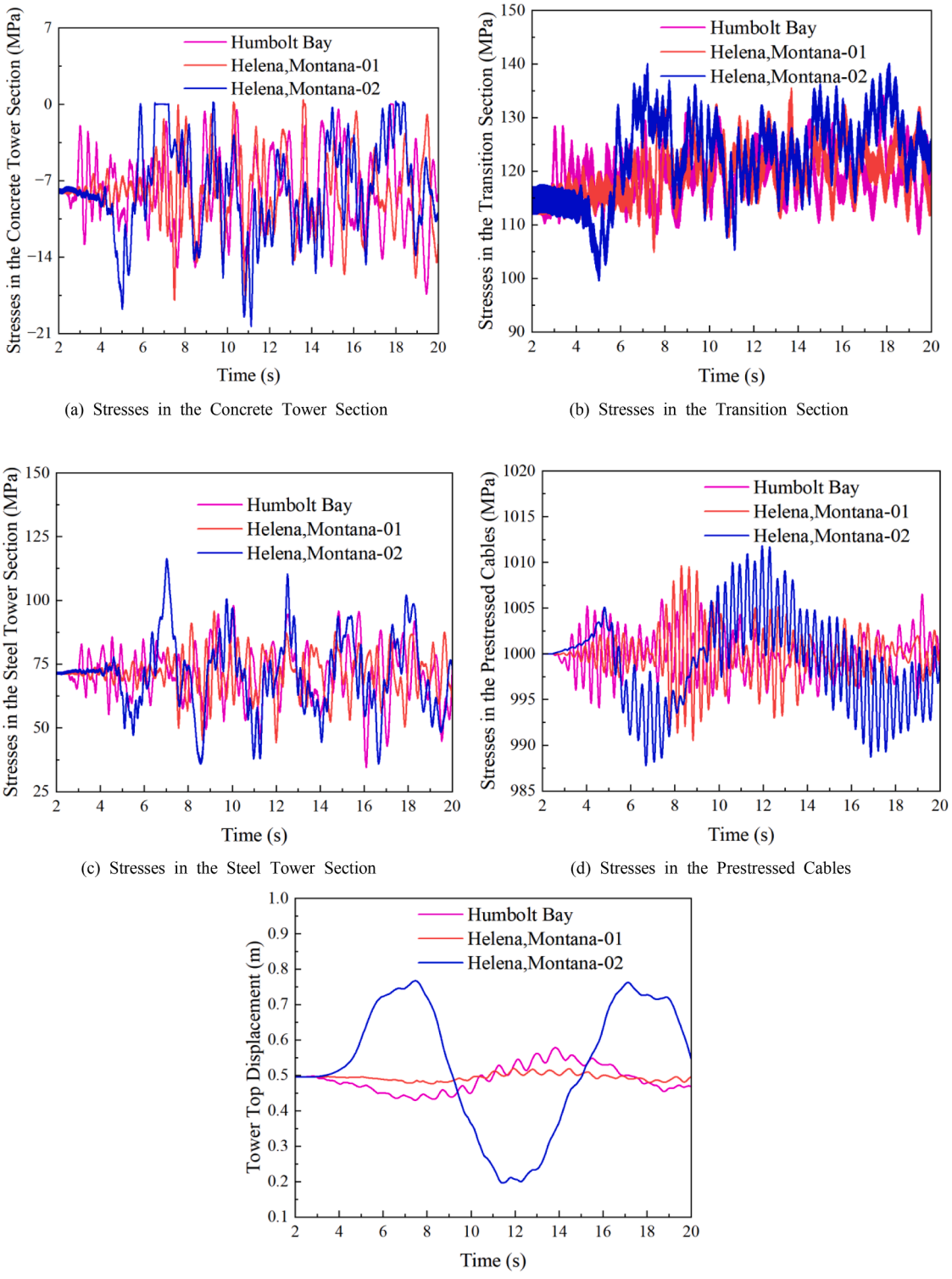
no occurrence of concrete tensile stress. The maximum concrete compressive stresses were 12.19 MPa, 12.50 MPa, and 13.60 MPa, respectively. These values are significantly lower than the design compressive strength of C80 concrete, indicating that there is no risk of tensile cracking in the concrete tower section under design earthquakes. Furthermore, the concrete compressive stress remains at a low level with a sufficient safety margin, thereby validating the reliability of the prestress design.

Regarding the steel components, the maximum stresses in the transition section under the three seismic waves were 125.49 MPa, 126.29 MPa, and 132.22 MPa, respectively. Similarly, the maximum stresses in the upper steel tower section were 83.33 MPa, 82.58 MPa, and 92.37 MPa, respectively. Both sets of values are significantly lower than the yield strength of Q345 steel. This demonstrates that the steel structure

components remain within the elastic range under design earthquakes, possessing a high seismic safety margin.

The initial tension of the prestressed cables is 1000 MPa. Under the specified seismic waves, the maximum stresses increased to 1002.76 MPa, 1003.92 MPa, and 1005.20 MPa, respectively. The stress increments were merely 0.28 %, 0.39 %, and 0.52 %, leaving a substantial margin relative to the ultimate strength of the steel strands. This indicates that the prestressed cables maintain a stable constraining effect under design earthquakes.

Tower top displacement serves as a critical indicator of overall structural stiffness and stability. Under the Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves, the tower top displacements were 0.53 m, 0.51 m, and 0.62 m, respectively, corresponding to 0.33 %, 0.31 %, and 0.38 % of the total tower height



(a) Stresses in the Concrete Tower Section

(b) Stresses in the Transition Section

(c) Stresses in the Steel Tower Section

(d) Stresses in the Prestressed Cables

(e) Tower Top Displacement

Fig. 11. Dynamic Response of the Hybrid Tower under Rare Earthquake.

(163.24 m). For such super-tall flexible structures, these deformations fall within conventional design limits, suggesting that the overall deformation of the hybrid tower is well-controlled under design earthquakes, posing no threat to operational safety or structural stability.

In summary, under design earthquake actions, the concrete tower section, the steel structure sections, and the prestressed cables of the hybrid tower all exhibit favorable mechanical performance, and the tower top displacement remains within a reasonable range. The overall disaster-resistance performance is excellent, ensuring the safety and normal operation of the structure under this condition.

5.3. Disaster-Resistance performance under rare earthquakes

The time histories of stresses in the concrete tower section, the transition section, the upper steel tower section, and the prestressed cables, as well as the tower top displacement time history under rare earthquakes, are presented in Fig. 11. The research reveals that under rare earthquakes, tensile stress emerges in the concrete tower section, posing a risk of dislocation and cracking in the concrete segments. Consequently, the safety and stability of the structure face severe challenges, indicating a marked insufficiency in disaster-resistance performance.

The most critical threat to the hybrid tower under rare earthquake action manifests in the concrete tower section. As illustrated in Fig. 11 (a), tensile stress occurred in the concrete tower section under the action of the Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves. For the concrete tower section, which relies on segmental assembly and prestressing, the emergence of tensile stress implies that the initial compressive stress provided by the prestressing has been completely counteracted by the bending moments induced by the earthquake. For the precast concrete tower section relying on segmental assembly, the emergence of tensile stress indicates that the initial pre-compression at the critical cross-section has been overcome by the seismic bending moments. Although the maximum tensile stress (0.40 MPa) is significantly lower than the tensile strength of C80 concrete (3.9 MPa) and will not cause material rupture, it signifies the onset of joint opening (decompression) at the segment interfaces. This opening can lead to a reduction in local stiffness and, under cyclic seismic loading, creates a risk of joint dislocation and durability issues, violating the "zero-tension" design criterion typically preferred for such critical prestressed structures.

In contrast to the concrete tower section, the steel structure components and prestressed cables remain within safe limits. Under the action of the Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves, the maximum stresses in the transition section were 134.13 MPa, 135.48 MPa, and 140.09 MPa, respectively, while the maximum stresses in the upper steel tower section were 97.90 MPa, 95.80 MPa, and 116.36 MPa, respectively. Although these stress values increased, they remain significantly lower than the 345 MPa yield strength of Q345 steel, indicating that the steel structure possesses ample safety reserves. Meanwhile, the maximum stresses in the prestressed cables increased to 1006.59 MPa, 1009.24 MPa, and 1010.43 MPa, respectively. Compared to the standard ultimate strength of 1860 MPa, the stress increments are minimal, maintaining a substantial safety margin.

Regarding overall structural deformation, the tower top displacement increased significantly. Under the action of the Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves, the tower top displacements were 0.58 m, 0.52 m, and 0.77 m, respectively, representing 0.36 %, 0.32 %, and 0.47 % of the tower height. Compared to the frequent earthquake and design earthquake conditions, the tower top displacement is markedly amplified, suggesting that the overall structural stiffness is compromised under rare earthquakes.

In summary, although the material stress levels remain within the elastic range, the current design does not fully satisfy the strict "full-compression" performance objective desirable for prestressed segmental towers under rare earthquakes. The appearance of tensile stress, while insufficient to crack the concrete material itself, introduces the risk of joint opening and potential stiffness degradation. To ensure optimal structural integrity and prevent cyclic joint damage, enhancement measures are necessary.

6. Measures to enhance the disaster-resistance performance of the hybrid tower

The preceding analysis indicates that the hybrid tower exhibits favorable disaster-resistance performance under strong wind conditions, frequent earthquakes, and design earthquakes. However, under rare earthquakes, tensile stress emerges in the concrete tower section, creating risks of segment joint dislocation and cracking. This insufficient overall disaster-resistance performance constitutes a weak link constraining structural safety. Consequently, it is necessary to propose feasible measures to enhance disaster-resistance performance based on the mechanical characteristics of the hybrid tower under rare earthquakes.

To enhance the disaster-resistance performance of the hybrid tower and eliminate potential safety hazards in the concrete tower section under rare earthquakes, increasing the pre-tension value of the prestressed cables is proposed as a solution. By increasing the prestress level, this study aims to prevent the occurrence of tensile stress in the concrete tower section while ensuring that the dynamic responses of other critical components and the tower top displacement remain within safe limits, thereby achieving a synergistic enhancement of the overall structural disaster-resistance performance.

In this section, the pre-tension value of the prestressed cables is increased from the original design value of 1000 MPa to 1219 MPa. On this basis, the dynamic responses of the hybrid tower under strong wind conditions and rare earthquakes are analyzed to evaluate the impact of increased prestress on disaster-resistance performance.

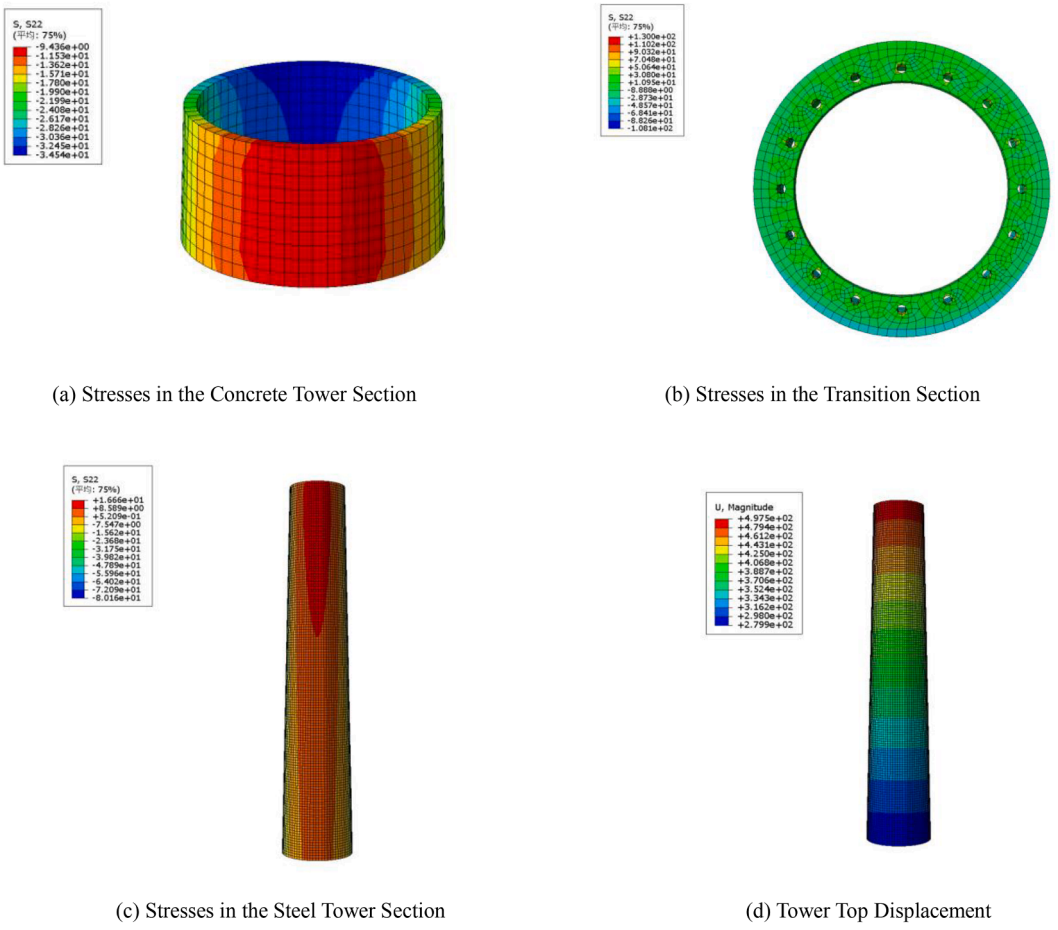
After the prestress was enhanced, under the combined actions of gravity and prestress only, the cloud diagrams of stresses in the concrete tower section, stresses in the transition section, stresses in the upper steel tower section, and stresses in the prestressed cables, together with the cloud diagram of tower top displacement, are shown in Fig. 12. From the mechanical performance of the concrete tower section, the maximum stress after prestress enhancement was 9.43 MPa, indicating a compressive state. No concrete tensile stress was observed, and the stress level was far below the design value of the compressive strength of C80 concrete. This demonstrates that, with increased prestress, the concrete tower section is not subjected to excessive concrete compressive stress, and thus the risk of crushing failure of the concrete tower section is avoided.

The stresses in the transition section and the upper steel tower section, as well as the tower top displacement, remained within reasonable ranges. The maximum stress in the transition section was 130 MPa, and the maximum stress in the upper steel tower section was 16.66 MPa; both values are lower than the yield strength of Q345 steel (345 MPa). The maximum tower top displacement was 0.49 m, accounting for 0.30 % of the tower height, which is also within the allowable safety limit.

In summary, with the enhanced prestress scheme, the hybrid tower under self-weight and prestress successfully eliminates the potential risk associated with concrete tensile stress in the concrete tower section, while avoiding excessive compressive stress that could lead to crushing. Overall, all components of the structure remain in a safe state.

6.1. Response under rare earthquake action with enhanced prestress

With enhanced prestress, the time histories of stresses in the concrete tower section, the transition section, the upper steel tower section, and the prestressed cables, as well as the tower top displacement time history under rare earthquakes, are presented in Fig. 13. Regarding the mechanical state of the concrete tower section, under the action of the



(a) Stresses in the Concrete Tower Section

(b) Stresses in the Transition Section

(c) Stresses in the Steel Tower Section

(d) Tower Top Displacement

Fig. 12. Cloud diagrams of stresses in the hybrid tower after prestress enhancement.

Humbolt Bay, Helena, Montana-01, and Helena, Montana-02 seismic waves, the section remained in a compressive state throughout the entire analysis period, with no recurrence of tensile stress. The corresponding maximum concrete compressive stresses were 17.92 MPa, 18.68 MPa, and 16.91 MPa, respectively. Although these values are higher than those in the original prestress condition (1000 MPa), they remain significantly lower than the design compressive strength of C80 concrete. This indicates that increasing prestress effectively reduces the risk of tensile cracking in the concrete tower section while maintaining a sufficient safety margin, thereby enhancing the safety reserve and structural stability of the concrete tower section.

The stresses in the transition section and the upper steel tower section, as well as the tower top displacement, remain within reasonable ranges. Under the action of the three seismic waves, the maximum stresses in the transition section were 163.43 MPa, 164.60 MPa, and 160.83 MPa, respectively. Although slightly increased due to the application of enhanced prestress, they remain lower than the yield strength of Q345 steel (345 MPa), indicating ample safety reserves. The maximum stresses in the upper steel tower section were 95.04 MPa, 93.20 MPa, and 83.80 MPa, respectively; these values are lower than those in the original condition, keeping the section in a safe state. Regarding structural deformation, the maximum tower top displacements were 0.57 m, 0.52 m, and 0.58 m, representing 0.35 %, 0.32 %, and 0.36 % of the tower height, respectively. The reduction in maximum tower top displacement compared to the original condition suggests that the increased prestress has, to a certain extent, enhanced the overall structural stiffness and mitigated deformation under seismic action.

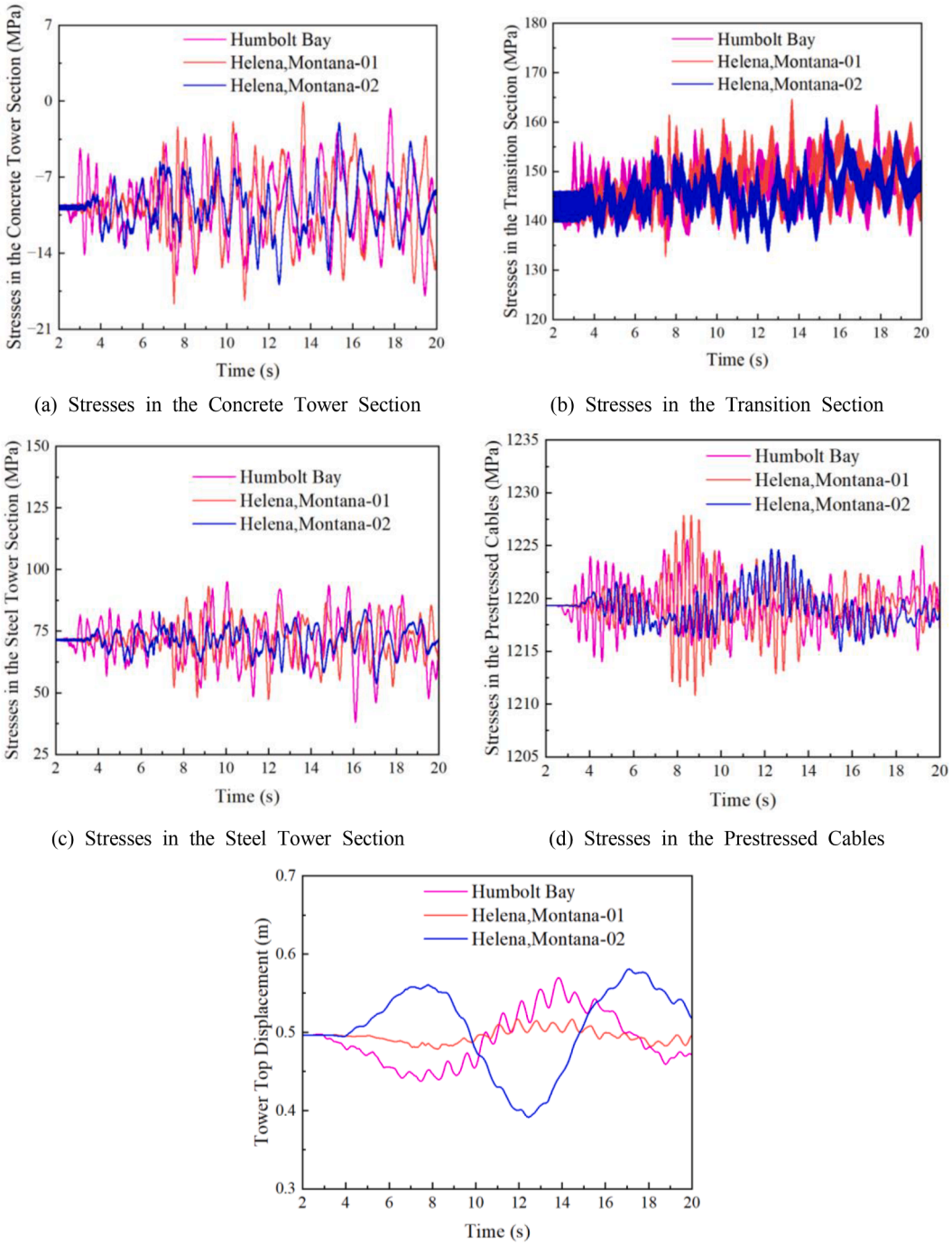
The pre-tension value of the prestressed cables was increased to 1219 MPa. Under rare earthquake action, the maximum stress increased to 1227.86 MPa. The stress increment was only 8.86 MPa, remaining far below the ultimate strength of the steel strands. This demonstrates that even after the pre-tension value is increased, the prestressed cables maintain stable performance and continuously provide effective constraint to the concrete tower section, thereby enhancing the disaster-resistance performance of the structure.

In summary, after adopting the measure of enhanced prestress, the hybrid tower exhibits favorable disaster-resistance performance under rare earthquakes. This measure successfully eliminates the hazard of tensile stress in the concrete tower section while ensuring all parts of the overall structure remain in a safe state. The analysis results indicate that enhancing prestress is an effective approach to improve the seismic performance of the hybrid tower, offering a solution to prevent collapse under rare earthquakes.

6.2. *Response under strong wind conditions with enhanced prestress*

The preceding analysis indicates that the hybrid tower possesses superior wind resistance under the original design prestress. To investigate the influence of the "enhanced prestress" scheme on the wind-induced response of the hybrid tower, a verification analysis was conducted under strong wind conditions with an applied pre-tension value of 1219 MPa. The time histories of stresses in the concrete tower section, the transition section, the upper steel tower section, and the prestressed cables, along with the tower top displacement time history, are presented in Fig. 14. The research reveals that the hybrid tower continues to exhibit overall favorable disaster-resistance performance under strong wind conditions.

Under strong wind conditions, the overall stress levels of the hybrid



(a) Stresses in the Concrete Tower Section

(b) Stresses in the Transition Section

(c) Stresses in the Steel Tower Section

(d) Stresses in the Prestressed Cables

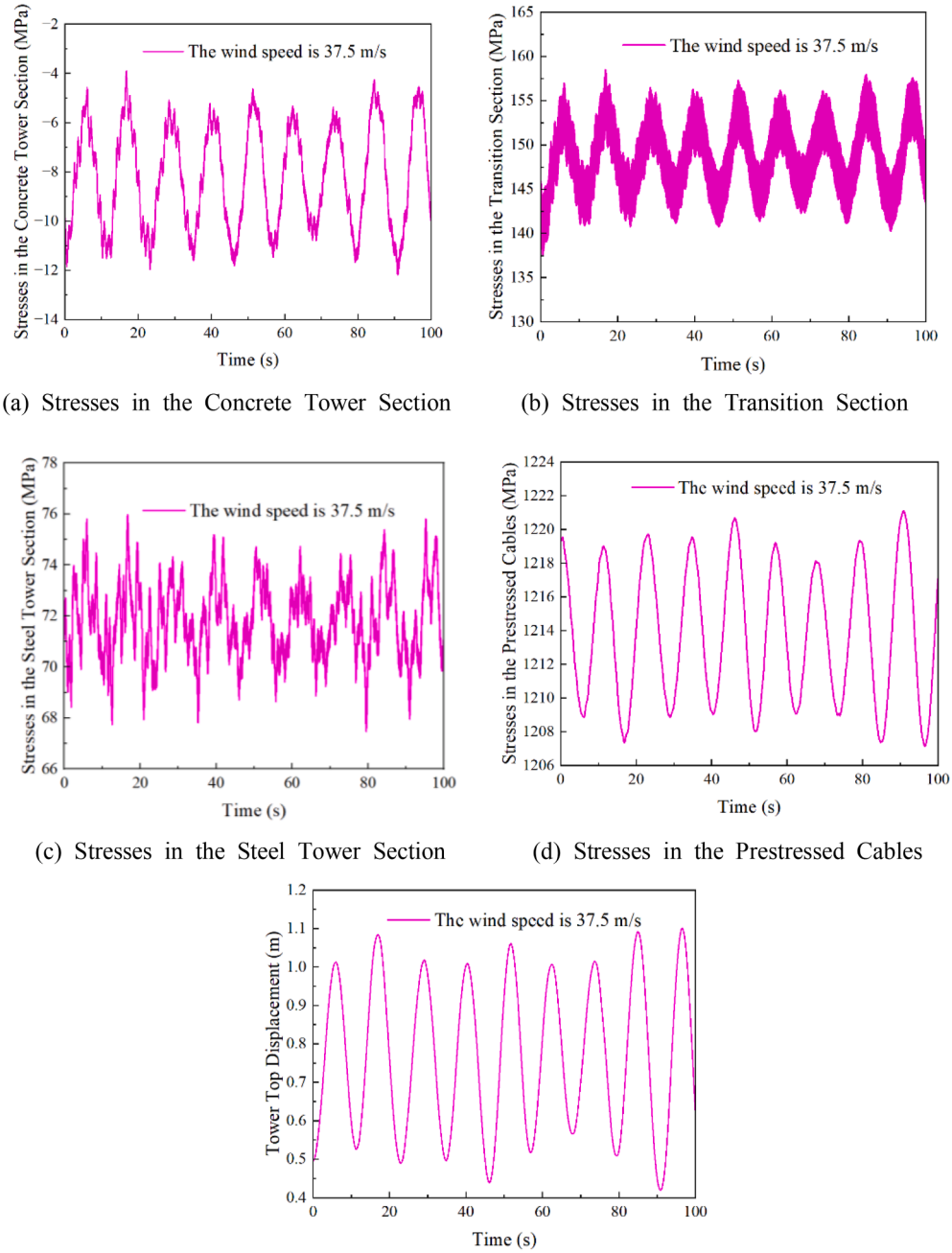
(e) Tower Top Displacement

Fig. 13. Dynamic response of the hybrid tower under rare earthquake.

tower with enhanced prestress remain within a safe range. As illustrated in Fig. 13(a), the concrete tower section exhibits compressive stress throughout the entire duration of the wind load, with no occurrence of tensile stress. The maximum concrete compressive stress is 12.2 MPa; although slightly lower than in the original condition, it remains significantly below the design compressive strength of C80 concrete. This indicates that the enhancement of prestress has not compromised the structural safety of the concrete tower section under high wind action; furthermore, due to the sufficient reserve of compressive stress, there is no risk of tensile cracking. The maximum stress in the upper steel tower section is 75.97 MPa, representing only a marginal increase, and the section remains in the elastic stage. The maximum stress in the prestressed cables under wind loads is 1221.09 MPa with a stress increment of only 2.09 MPa, demonstrating that the prestressed cables maintain stability under high wind action, continuously providing effective constraint and ensuring structural integrity.

Following the increase in prestress, the maximum tower top displacement increased to 1.1 m, representing approximately 0.67 % of the total tower height. This increase is mainly attributed to the higher axial compression introduced by enhanced prestress, which amplifies the second-order (P–Δ) effect under strong wind; nevertheless, the displacement remains within conventional design limits and does not threaten the normal operation or structural safety of the hybrid tower.

In summary, after enhancing the prestress, the hybrid tower



(a) Stresses in the Concrete Tower Section

(b) Stresses in the Transition Section

(c) Stresses in the Steel Tower Section

(d) Stresses in the Prestressed Cables

(e) Tower Top Displacement

Fig. 14. Dynamic response of the hybrid tower under wind hazard.

continues to demonstrate excellent disaster-resistance performance under strong wind conditions. Stresses in all critical components are significantly lower than material strength limits, and the overall structural displacement is controlled within a reasonable range. Comparative analysis indicates that the trend of structural dynamic response under strong wind conditions aligns well with that observed under rare earthquake conditions. This compellingly proves that the "enhanced prestress" method for improving disaster-resistance performance possesses excellent applicability and safety across different hazard conditions.

7. Conclusions

Taking a 160 m-class hybrid tower located in a region with a seismic fortification intensity of 8 as the prototype, this study established a refined finite element model using ABAQUS software. Dynamic time-history analyses were conducted under strong wind conditions, frequent earthquakes, design earthquakes, and rare earthquakes to investigate the responses of the concrete tower section, the steel structure components, and the prestressed cables. The following conclusions are drawn:

(1) The hybrid tower exhibits favorable mechanical performance and remains in a safe state under strong wind conditions. Under a basic wind speed of 37.5 m/s, the stress levels in all components are lower than the design material strengths, providing sufficient safety margins. The maximum tower top displacement is 0.57 m,

representing 0.35 % of the total height, which falls within the safe range.

(2) Under frequent earthquakes and design earthquakes, the hybrid tower demonstrates favorable disaster-resistance performance. Under these two types of seismic action, no tensile stress occurs in the concrete tower section. The stress levels in critical components—such as the steel structure and prestressed cables—and the tower top displacement remain within allowable design limits. This indicates that the hybrid tower maintains good stability and safety under frequent and design earthquakes.

(3) Under rare earthquake action, the disaster-resistance performance of the hybrid tower is insufficient, presenting a risk of collapse. Tensile stress emerges in the concrete tower section, reaching a maximum value of 0.40 MPa. For the concrete tower section, the emergence of tensile stress indicates that the pre-compression stress at the concrete segment joint has been completely offset. This implies a potential opening of the segmented joints throughout the tower, thereby increasing the risk of collapse.

(4) Increasing the pre-tension value of the prestressed cables by 21.9 % effectively ensures the safety of the structure under rare earthquake action. With the increased pre-tension value, tensile stress is eliminated in the concrete tower section, and the risks of segment joint dislocation and cracking are effectively controlled. Furthermore, the stresses in the steel structure and prestressed cables, as well as the tower top displacement, remain within safe ranges. Simultaneously, under strong wind conditions, all components of the hybrid tower maintain their safety. This demonstrates that enhancing the pre-tension value is an effective measure to improve the disaster-resistance performance of the hybrid tower.

Relevance

This study demonstrates that the prestress level is a critical parameter determining the life-cycle disaster-resistance resilience of the hybrid tower. Simulation results indicate that the hybrid tower maintains a sufficient safety margin under an initial tensile stress of 1000 MPa in the prestressed cables. Specifically, under a basic wind speed of 37.5 m/s combined with frequent earthquake or design earthquake actions, the structure remains in an elastic working state, with both the tower top displacement and component stresses staying within safe limits. However, under rare earthquake conditions, the structure exhibits a significant deficiency in resilience, characterized by a maximum tensile stress of 0.40 MPa in the concrete tower section. This indicates the occurrence of openings at the segment joints within the concrete tower section, which increases the risk of collapse and diminishes the seismic resilience of the hybrid tower. To address this vulnerability, the study confirms that increasing the cable prestress by 20 % is an effective measure for enhancing structural resilience. This adjustment successfully counteracts the tensile stress in the concrete tower section under rare earthquake conditions while ensuring that the compressive stress retains a high safety margin, thereby mitigating the risk of joint opening. Furthermore, after increasing the tensile stress of the prestressed cables, the responses of the steel structure, the stresses in the prestressed cables, and the tower top displacement remain within safe limits. Consequently, increasing the cable prestress serves as an effective method to improve the disaster-resistance resilience of the hybrid tower.

CRediT authorship contribution statement

Wenming Wang: Writing – original draft, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Bohai Ji:** Writing – review & editing, Validation, Supervision, Software, Resources. **Huichao Ma:** Writing – review & editing, Visualization, Validation, Supervision. **Han Diao:** Visualization, Software, Resources. **Zhenli Zhang:** Visualization, Validation.

Disclosure statement

The authors report there are no competing interests to declare.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.rcns.2026.03.002.

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